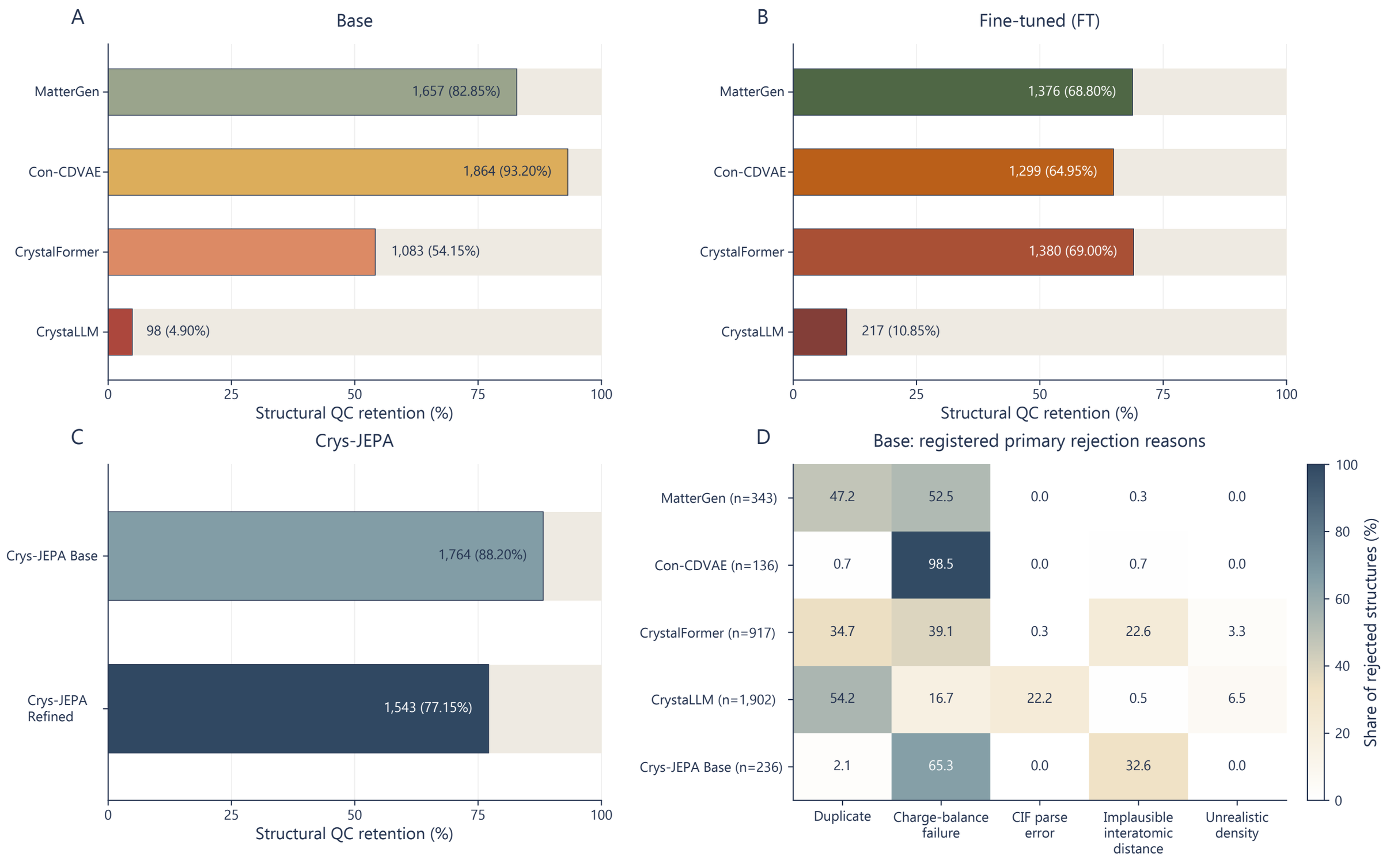

A
Base
MatterGen
Con-CDVAE
CrystalFormer
CrystaLLM
1,657 (82.85%)
1,864 (93.20%)
1,083 (54.15%)
98 (4.90%)
0
25
50
75
100
Structural QC retention (%)
B
Fine-tuned (FT)
MatterGen
Con-CDVAE
CrystalFormer
CrystaLLM
1,376 (68.80%)
1,299 (64.95%)
1,380 (69.00%)
217 (10.85%)
0
25
50
75
100
Structural QC retention (%)
C
Crys-JEPA
Crys-JEPA Base
Crys-JEPA Refined
1,764 (88.20%)
1,543 (77.15%)
0
25
50
75
100
Structural QC retention (%)
D
Base: registered primary rejection reasons
MatterGen (n=343)
Con-CDVAE (n=136)
CrystalFormer (n=917)
CrystaLLM (n=1,902)
Crys-JEPA Base (n=236)
47.2
52.5
0.0
0.3
0.0
0.7
98.5
0.0
0.7
0.0
34.7
39.1
0.3
22.6
3.3
54.2
16.7
22.2
0.5
6.5
2.1
65.3
0.0
32.6
0.0
Duplicate
Charge-balance failure
CIF parse error
Implausible interatomic distance
Unrealistic density
Share of rejected structures (%)
0
20
40
60
80
100